

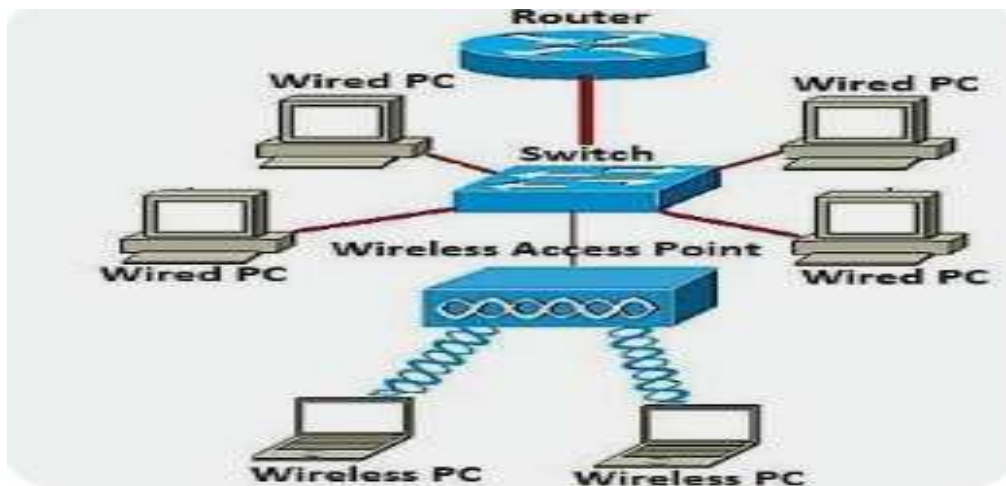
## Question : 1

Draw a network topology diagram for a small café Wi-Fi setup using the star topology, labeling all devices (router, access points, user devices) and connections clearly.

## Answer: 1

### Small Café Wi-Fi Setup Layout (Star Topology)

A classic **Star Topology** uses a central node (the core switch) to link all peripheral hardware nodes. In this layout, if a single cable breaks or a smartphone disconnects, the rest of the café network continues running smoothly without interruption.



### Topology Component Mapping & Connections

- **Core Internet Gateway:**
  - **Internet (ISP Modems):** Connects directly to the **Core Router** using a WAN Ethernet cable interface.
- **The Central Hub (Star Core):**
  - **Central Switch:** Connects directly to the router via a LAN uplink cable. This switch functions as the main central node of the physical star layout.
- **Wired Star Extensions (Spokes):**
  - **Wireless Access Points (APs):** Multiple ceiling-mounted Access Points connect directly back to the central switch using individual **PoE (Power over Ethernet) Cables**.
  - **Wired Café Devices:** Hardware assets like the **POS Billing Desktop** and the **Network Kitchen Printer** are wired directly to dedicated ports on the central switch.
- **Wireless Client Endpoints:**
  - **User Devices:** Customer smartphones, customer laptops, and café order-taking tablets connect wirelessly over the **Wi-Fi Radio Waves** broadcast by the individual Access Points, forming a clean local wireless distribution star cluster.

## Question: 2

Design a network for a two-floor co-working space (like a mini WeWork) using a ring topology for each floor and explain how devices are connected to maintain communication between floors.<br><br><em><strong>Hint:</strong> Show how the ring topology helps in handling device failures on one floor without affecting the other.</em>

## Answer: 2

### Two-Floor Co-Working Space Network Design (Ring Topology)

In this setup, each floor operates on its own dedicated physical ring network, and both floors are linked together through a central core layer to ensure smooth communication.

#### Floor 1 Connections:

All user workstations, printers, and switches on the first floor are connected sequentially in a closed loop using fiber-optic cables. Device A connects to Device B, Device B connects to Device C, and the last device loops back to connect directly to Device A.

#### Floor 2 Connections:

The second floor uses the exact same layout. All local workspace nodes are connected in a separate, independent ring loop.

#### Inter-Floor Communication:

To allow the two floors to talk to each other, one primary switch from the Floor 1 ring and one from the Floor 2 ring are designated as "Bridge Nodes." Both of these switches are wired directly to a Central Layer 3 Core Switch (or Router) located in the main server room. When a user on Floor 1 sends data to a printer on Floor 2, the traffic travels along the Floor 1 ring, moves up through the Core Switch, and is routed directly into the Floor 2 ring.

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### How it Handles Device Failures

**Fault Isolation Between Floors:** Because the rings are physically separated on each floor, a total network crash or a broken cable on Floor 1 stays isolated. The ring on Floor 2 will keep running perfectly without any downtime.

**Redundant Ring Protection:** If you use a Dual-Ring Topology (FDDI style) on a floor, the network can survive a node failure. If one laptop or switch goes offline, the ring instantly reverses the traffic path in the opposite direction, keeping the rest of that floor online while the broken device is fixed.

### Question: 3

Create a mesh topology diagram for a gaming café with 6 PCs, showing how redundancy is achieved if any one PC or cable fails

### Answer: 3

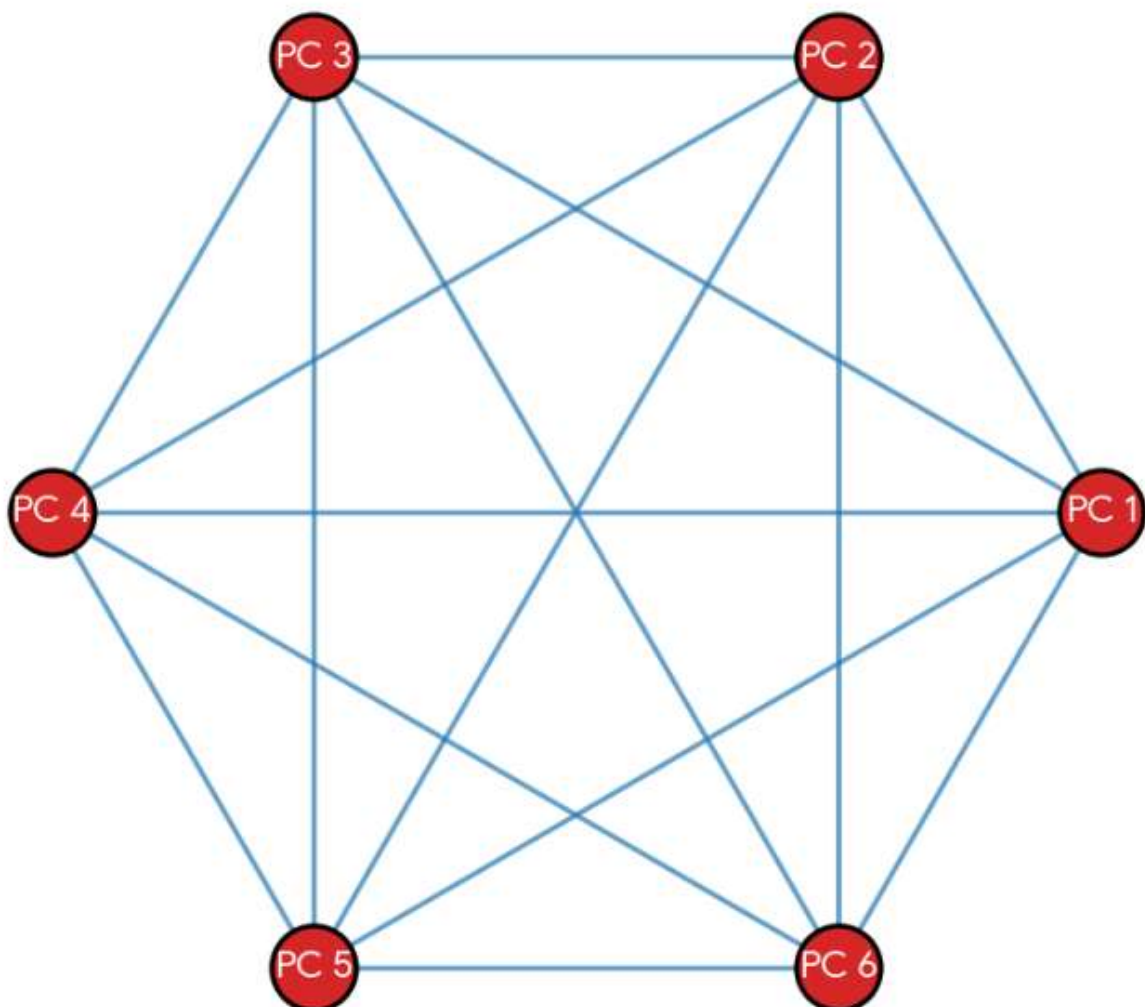
#### 6-PC Mesh Topology Layout

In a **Full Mesh Topology**, every single computer connects directly to every other computer in the room. For a network with **6 PCs**, the total number of dedicated cables needed is calculated using the formula:

$$\text{Total Links} = \frac{n(n-1)}{2} = \frac{6 \times 5}{2} = 15 \text{ links}$$

As requested, the logical network mapping diagram demonstrating how these 6 gaming PCs interconnect to achieve total network redundancy is shown below:

#### Full Mesh Network Layout (6 PCs – 15 Links)



## How Redundancy is Achieved During Failures

- **Instant Backup Routing:** If the direct Ethernet cable running between **PC 1** and **PC 4** gets damaged or fails completely, connection downtime is instantly avoided. The network automatically reroutes the active data packets through any other online neighbor node (for example, sending the game data from **PC 1 → PC 2 → PC 4**).
- **Fault Isolation:** The failure of an individual cable or network interface card affects only that specific direct path. Because there is no single central switch or router controlling the interconnects, the remaining 5 gaming PCs stay online and can continue communicating without experiencing lag or system disconnects.

## Question: 4

Plan a multi-site network for a food delivery company with 3 city offices (Ahmedabad, Surat, Vadodara) — show how you would connect the sites, what topology you would use between them, and briefly explain your choice.

## Answer: 4

### Multi-Site Network Plan for Food Delivery Company

To interconnect the three city offices (**Ahmedabad**, **Surat**, and **Vadodara**), the infrastructure layout is designed using a **Full Mesh Topology** at the wide-area network (WAN) level.

text

[ Ahmedabad Office ]

/      \

(WAN Link)   (WAN Link)

/      \

[ Surat Office ] ---- [ Vadodara Office ]

(WAN Link)

- **How the Sites are Connected:**  
Each city office is equipped with a dedicated enterprise **VPN Gateway Router**. The offices are interconnected over secure, high-speed **Site-to-Site IPsec VPN tunnels** running on dedicated business broadband lines. This creates an encrypted private network overlay across the public internet.

- **Topology Choice: Full Mesh Topology**

Every city office maintains a direct point-to-point network connection to the other two offices. Ahmedabad connects directly to Surat and Vadodara, while Surat and Vadodara maintain a direct link between themselves.

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## Justification for My Choice

- **Zero Single Point of Failure:** For a live food delivery company, system downtime means lost restaurant orders and delivery delays. If the primary direct link between Ahmedabad and Surat fails due to an ISP outage, the routers will instantly reroute live order data through the Vadodara office (**Ahmedabad → Vadodara → Surat**) with zero system interruption.
- **Low Latency for Live Updates:** A mesh topology eliminates the need for a central hub office. Traffic like live driver tracking coordinates or order assignments can travel directly between any two cities using the shortest route, ensuring fast performance and minimal lag.

## Question: 5

Imagine your network needs to double in size next year (e.g., from 10 to 20 devices per site). Write a short expansion plan describing what changes you would make to your original topology to ensure scalability and minimal downtime.<br><br><em><strong>Hint:</strong> Consider adding switches, routers, or changing topology types if needed.</em>

## Answer: 5

### Network Expansion & Scalability Plan

To double the network size from **10 to 20 devices per site** next year while keeping network downtime near zero, the following upgrades will be implemented:

- **Hardware Upgrades for Scalability:**  
The current 16-port switches at each site will be upgraded by adding a second **24-port Gigabit Switch** (such as the Cisco Business 250 or TP-Link TL-SG1024D). The two switches will be connected via a high-speed **Gigabit Trunk Link** using an Link Aggregation Control Protocol (**LACP Port Channel**). This setup provides double the physical ports and prevents data bottlenecks between the switches.
- **Topology Evolution:**  
The local office layout will stick with a **Star Topology**, but it will expand into an **Extended Star Topology**. New employee workstations will be split evenly across both switches. This strategy ensures that if one switch fails, only half the office goes offline, rather than the entire site crashing.

- **Minimizing Expansion Downtime:**

To prevent business disruptions, all new hardware configurations, IP address updates, and cable management runs will be set up and pre-tested after business hours. Shifting the active network connections over to the new switches will take less than 10 minutes during a scheduled weekend maintenance window.